

Control work 4, and work on the mistakes

Every solid of the year in one paper, then a named diagnosis of each slip.

LESSON 59–60

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 4

IGX 13 REVIEW

LESSON CLOCK

3

45'

12'

20'

5'

Instructions	3 min
The paper	45 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Apply the surface-area and volume formulas of the whole year in one assessment.
- Handle compound solids, bounds and context correctly.
- Classify each lost mark and rewrite the solution in full.
- Draw the year’s mensuration as a single map.

TEXTBOOK

National	pp. 311–314
Cambridge	Core & Extended, pp. 393–396
Workbook	Control paper G4

01

Explanation

The paper — 40 marks, 45 minutes

Q	TASK	MARKS	FROM
1	A regular hexagonal prism, edge 4, height 9: total area and volume	6	Q II
2	A square pyramid, base edge 12, height 8: apothem, lateral area, volume	7	Q III
3	A cone of radius 6 and height 8: slant, curved area, volume, sector angle of the net	7	Q III
4	A ball of volume 972π : radius and surface area; then a cap of height 3	6	Q III
5	A silo — cylinder $r = 3$, $h = 8$ m plus a hemisphere: volume and painted area, base excluded	7	Q III–IV
6	A closed cylindrical tin of 500 cm^3 : the radius of least metal, verified	7	Q IV

WHERE THE MARKS ACTUALLY GO

Q2 carries one mark for using the apothem in the area and the height in the volume; Q3 one for the sector angle; Q5 one for excluding the covered circles; Q6 one for verifying the minimum. Four of the forty marks are for one decision each.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
slant for height	$V = \frac{1}{3}Bl$	volume uses the perpendicular
height for slant	$S = \frac{1}{2}Ph$	area uses the apothem
missing $\frac{1}{3}$	$V = Bh$	pyramids and cones are a third
sphere formulas swapped	$S = \frac{4}{3}\pi R^3$	S has R^2 , V has R^3
joined faces counted	adding two whole surface areas	subtract twice the shared face
ratios not cubed	$V_1 : V_2 = k$	k, k^2, k^3
minimum not verified	$f' = 0$	add $f'' > 0$

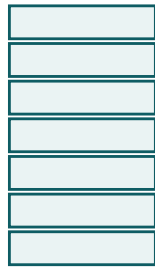
Write the slip's name in the margin of every question that lost a mark, then rewrite the whole solution — not the wrong line.

The year as one map

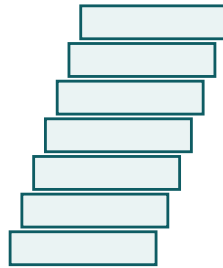
Eight boxes, links written as sentences:

- **coordinates in space** → **vectors** — “a point becomes three numbers”
- **vectors** → **the scalar product** — “angles without a picture”
- **prism** → $V = Bh$ — “uniform cross-section”
- **cylinder** → **the same, with $B = \pi r^2$**
- **pyramid and cone** → **a third of it** — “the same base, the same height”
- **sphere** → **2 : 3 with its cylinder** — “Archimedes, for both”
- **compound** → **add or subtract, and drop the joined faces**
- **scale k** → k, k^2, k^3 — “length, area, volume”

same base, same height, same volume



right prism



oblique prism

every horizontal slice has the same area

Equal slices, equal volumes — the principle under half the boxes above.

NINE FORMULAS, ONE PAGE

$$V_{prism} = Bh \quad V_{cyl} = \pi r^2 h \quad V_{pyr} = \frac{1}{3} Bh \quad V_{cone} = \frac{1}{3} \pi r^2 h \quad V_{sph} = \frac{4}{3} \pi R^3$$

$$S_{cyl} = 2\pi r(r + h) \quad S_{cone} = \pi r(r + l) \quad S_{sph} = 4\pi R^2 \quad V_{frus} = \frac{\pi h}{3} (R^2 + Rr + r^2)$$

If these nine sit on one page in your own handwriting, the annual review needs nothing new.

Looking forward

Lessons 61–68 are pure revision: coordinates and vectors, parallelism and perpendicularity, all the surfaces and volumes, then the entrance paper and the annual review. Nothing is added; the work is to make what exists fast and certain.

ONE HABIT TO CARRY FORWARD

The whole year rewarded writing B , h and the shape's name before any arithmetic. Keep that habit into the entrance paper, where the question rarely says which solid it is describing and the first line of the answer has to.

02 Worked examples

EXAMPLE 1 Model answer, Q2: a square pyramid, base edge 12, height 8.

① Base apothem 6; $m = \sqrt{64 + 36} = 10$.

The first right triangle.

② $S_{lat} = \frac{1}{2}(48)(10) = 240$

Apothem, not height.

③ $B = 144$, so $S_{total} = 384$.

④ $V = \frac{1}{3}(144)(8) = 384$

Height, not apothem.

ANSWER

$$m = 10, S_{lat} = 240, V = 384$$

EXAMPLE 2 Model answer, Q3: a cone of radius 6 and height 8.

① $l = \sqrt{36 + 64} = 10$

② $S_{curved} = \pi(6)(10) = 60\pi$

③ $V = \frac{1}{3}\pi(36)(8) = 96\pi$

④ $\theta = 360^\circ \times \frac{6}{10} = 216^\circ$

ANSWER

$$l = 10, 60\pi, 96\pi, 216^\circ$$

EXAMPLE 3 Model answer, Q5: a silo, cylinder $r = 3$, $h = 8$ m with a hemisphere, painted above ground.

$$\textcircled{1} \quad V = \pi(9)(8) + \frac{2}{3}\pi(27) = 72\pi + 18\pi = 90\pi$$

$$\approx 283 \text{ m}^3.$$

$$\textcircled{2} \quad \text{Curved cylinder } 2\pi(3)(8) = 48\pi.$$

$$\textcircled{3} \quad \text{Hemisphere } 2\pi(9) = 18\pi.$$

The two joined circles do not count.

$$\textcircled{4} \quad S = 66\pi \approx 207 \text{ m}^2.$$

Base excluded.

ANSWER

$$90\pi \approx 283 \text{ m}^3; 66\pi \approx 207 \text{ m}^2$$

EXAMPLE 4 Model answer, Q6: a closed tin of 500 cm^3 using least metal.

$$\textcircled{1} \quad S = 2\pi r^2 + \frac{1000}{r}$$

$$\textcircled{2} \quad \frac{dS}{dr} = 4\pi r - \frac{1000}{r^2} = 0$$

$$\textcircled{3} \quad r^3 = \frac{250}{\pi} \Rightarrow r \approx 4.30$$

$$\textcircled{4} \quad S'' > 0, \text{ so a minimum; } h \approx 8.60 = 2r.$$

ANSWER

$$r \approx 4.30 \text{ cm}, h \approx 8.60 \text{ cm}$$

03 Interactive model

Work Q5 twice — once counting all the faces, once dropping the joined ones — and let the class name the difference.

The year in twelve questions

Volume of a prism:

$$Ph$$

$$Bh$$

$$\frac{1}{3} Bh$$

$$2Bh$$

Volume of a cylinder:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$\frac{1}{3} \pi r^2 h$$

Volume of a pyramid:

$$Bh$$

$$\frac{1}{3} Bh$$

$$\frac{1}{2} Bh$$

$$3Bh$$

Volume of a cone:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\pi r l$$

$$\frac{1}{3} \pi r^2 l$$

Lateral area of a regular pyramid:

$$\frac{1}{2} Ph$$

$$\frac{1}{2} Pm$$

$$Pm$$

$$Bm$$

Curved area of a cone:

$$\pi r^2$$

$$\pi r l$$

$$\pi r h$$

$$2\pi r l$$

Volume of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\pi R^3$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3}(R^2 + r^2)$$

$$\frac{\pi h}{3}(R^2 + Rr + r^2)$$

$$\pi h(R^2 - r^2)$$

$$\frac{\pi h}{2}(R + r)$$

Joined faces of a compound solid:

count twice

count once

do not count

are halved

Scale factor k multiplies area by:

k

k^2

k^3

$2k$

Least-metal closed cylinder:

$$h = r$$

$$h = 2r$$

$$h = \frac{r}{2}$$

$$h = 3r$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Prism	Prizma	Призма
Pyramid	Piramida	Пирамида
Cylinder	Silindr	Цилиндр
Cone	Konus	Конус
Sphere	Sfera	Сфера
Compound solid	Murakkab jism	Составное тело
Optimisation	Optimallashtirish	Оптимизация

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q2 needs the apothem for:

the volume

the lateral area

the base

nothing

Q5 excludes:

the curved surfaces

the two joined circles and the base

nothing

the hemisphere

Q6 needs, beyond $f' = 0$:

nothing

a check that it is a minimum

a second constraint

a sketch

Lessons 61–68 are:

new material

revision and the entrance paper

a project

a test only

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume of a prism, $B = 15$, $h = 8$

ANSWER 120

2. Volume of a cylinder, $r = 2$, $h = 7$

ANSWER 28π

3. Volume of a pyramid, $B = 144$, $h = 8$

ANSWER 384

4. Volume of a cone, $r = 6$, $h = 8$

ANSWER 96π

5. Slant of that cone

ANSWER 10

6. Radius of a ball of volume 972π

ANSWER 9

7. Surface area of that ball

ANSWER 324π

Medium · the standard question

7 PROBLEMS

1. Hexagonal prism, edge 4, height 9: volume

ANSWER $216\sqrt{3} \approx 374$

2. Same prism: total area

ANSWER $48\sqrt{3} + 216 \approx 299$

3. Square pyramid, base 12, height 8: apothem and lateral area

ANSWER 10, 240

4. Same pyramid: volume

ANSWER 384

5. Cone $r = 6$, $h = 8$: curved area and sector angle

ANSWER 60π , 216°

6. Cap of height 3 on a ball of radius 9: curved area

ANSWER 54π

7. Silo, cylinder $r = 3$, $h = 8$ plus hemisphere: volume

ANSWER $90\pi \approx 283$

Hard · stretch and reason

7 PROBLEMS

1. Same silo: painted area, base excluded

ANSWER $66\pi \approx 207 \text{ m}^2$

2. Closed tin of 500 cm^3 : the radius of least metal

ANSWER $\approx 4.30 \text{ cm}$

3. Same tin: the least area

ANSWER $\approx 349 \text{ cm}^2$

4. Cap of height 3 on a ball of radius 9: its volume

ANSWER $\frac{\pi}{3}(9)(27 - 3) = 72\pi$

5. Two similar cones of volumes 27 and 64: the area ratio

ANSWER $9 : 16$

6. A cone and a hemisphere on the same base of radius r have equal volumes: find h

ANSWER $h = 2r$

7. A frustum of radii R and $\frac{R}{2}$: the fraction of the full cone

ANSWER $\frac{7}{8}$

07 Homework

Homework — 4 tasks

Bring the nine-formula page and the concept map to Lesson 61.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Write the nine formulas of the year on one page in your own hand, each with a small sketch.
3. A grain silo is a cylinder of radius 2.5 m and height 9 m with a hemispherical top. Find its volume and the area painted above ground.

4. A closed cylindrical tin holds 750 cm^3 . Find the radius that uses least metal, and verify that it is a minimum.